

DATA-260 HOMEWORK 1

Sanjana Thummalapalli

PERSONAL CONFIGURATION

|                    |                                                                                                                     |
|--------------------|---------------------------------------------------------------------------------------------------------------------|
| Student Name       | Sanjana Thummalapalli                                                                                               |
| SID4               | 7801                                                                                                                |
| PORT_BASE          | 8601                                                                                                                |
| PREFIX             | s7801                                                                                                               |
| SEED               | 7801                                                                                                                |
| VERIFY_SEED        | 267801                                                                                                              |
| DOMAIN_ID          | 1                                                                                                                   |
| Assigned Domain    | Clinical Trial Listings                                                                                             |
| Python Version     | 3.11                                                                                                                |
| Local Model        | qwen3:4b                                                                                                            |
| Tagged Commit Hash | 6af9deedb359ee8c24f16cf7c671d770938e95fd                                                                            |
| GitHub Link        | <a href="https://github.com/sanjana-glitch-art/data260-7801">https://github.com/sanjana-glitch-art/data260-7801</a> |

HARDWARE

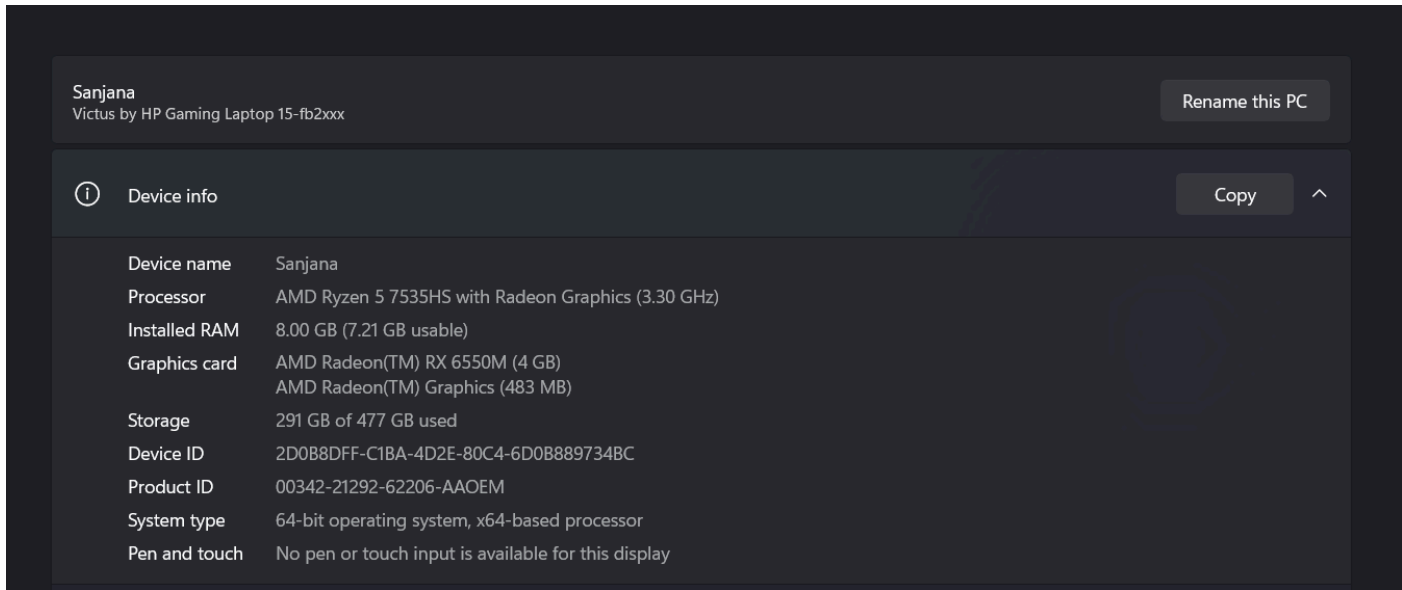


FIGURE 1: HARDWARE INFORMATION

MODEL SELECTION

I initially tested the requested qwen3:8b model. On my computer, one Planner-Reviewer pipeline run required approximately 519 seconds. My computer has 8 GB of RAM and a 4 GB AMD Radeon RX 6550M GPU. Because this runtime made the required 40-run experiment impractical, I used the smaller tool-capable qwen3:4b model.

PART 1 - HTML AND JAVASCRIPT

1. DOMAIN SCHEMA

Before implementing the HTML form, I defined the clinical-trial entity in DOMAIN\_SCHEMA.md. The primary field is trialTitle. The secondary field is sponsorName. The content field is trialDescription. The four category values are Phase I, Phase II, Phase III, and Phase IV.

Schema:

| Field            | Type          | Required  | Description                          |
|------------------|---------------|-----------|--------------------------------------|
| trialTitle       | Text          | Yes       | Official or descriptive trial title  |
| sponsorName      | Text          | Yes       | Email address of the submitter       |
| submitterEmail   | Email         | Yes       | Email address of the submitter       |
| trialDescription | Text ares     | Yes       | Description and purpose of the trial |
| trialPhase       | Category      | Yes       | Phase I, II, III, or IV              |
| termsAccepted    | Boolean       | Yes       | Terms and conditions acceptance      |
| submissionDate   | Date and time | Generated | Successful submission timestamps     |

## 2. HTML FORM

```

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>HW1-Sanjana Thummalapalli</title>

  <style> ...
</style>
</head>

<body>
  <h1>Clinical Trial Listing</h1>

  <form id="clinicalTrialForm">
    <label for="trialTitle">Trial title</label>
    <input
      type="text"
      id="trialTitle"
      name="trialTitle"
      placeholder="Enter the clinical trial title"
      required
      autofocus
    >

    <label for="sponsorName">Sponsor name</label>
    <input
      type="text"
      id="sponsorName"
      name="sponsorName"
      placeholder="Enter the sponsoring organization"
      required
    >

    <label for="submitterEmail">Submitter email</label>
    <input
      type="email"
      id="submitterEmail"
      name="submitterEmail"
      placeholder="Enter the submitter's email address"
      required
    >

    <label for="trialDescription">Trial description</label>
    <textarea
      id="trialDescription"
      name="trialDescription"
      placeholder="Describe the clinical trial, its purpose, and its participants"
    >
  </form>

```

**FIGURE 2: RELEVANT HTML FORM CODE**

**Clinical Trial Listing**

**Trial title**  
Enter the clinical trial title

**Sponsor name**  
Enter the sponsoring organization

**Submitter email**  
Enter the submitter's email address

**Trial description**  
Describe the clinical trial, its purpose, and its participants

**Trial phase**  
Select a trial phase

☐ I agree to the terms and conditions.

Submit Clinical Trial

**FIGURE 3: FORM RUNNING AT <http://localhost:8601>**

### 3. JAVASCRIPT VALIDATION

localhost:8601 says  
The trial description must contain more than 25 characters.

OK

Trial title  
asdfigasdf

Sponsor name  
dfgh

Submitter email  
s@dfg

Trial description  
erth

Trial phase  
Phase I

☒ I agree to the terms and conditions.

Submit Clinical Trial

**FIGURE 4: DESCRIPTION VALIDATION ALERT**

asdfig

**Submitter email**  
s@sdfg

**Trial description**  
zxcvsdfghjkl;ijkinhbj

Code  
You must agree to the terms and conditions.

OK

**Trial phase**  
Phase I

☐ I agree to the terms and conditions.

Submit Clinical Trial

**FIGURE 5: TERMS VALIDATION ALERT**

### 4. JSON, DESTRUCTURING, SPREAD OPERATOR, AND CLOSURE

After validation succeeds, the form data is converted to a JSON string and logged. The JSON string is parsed back into an object. Object destructuring extracts trialTitle and submitterEmail. The spread operator adds submissionDate. A closure tracks successful submissions.

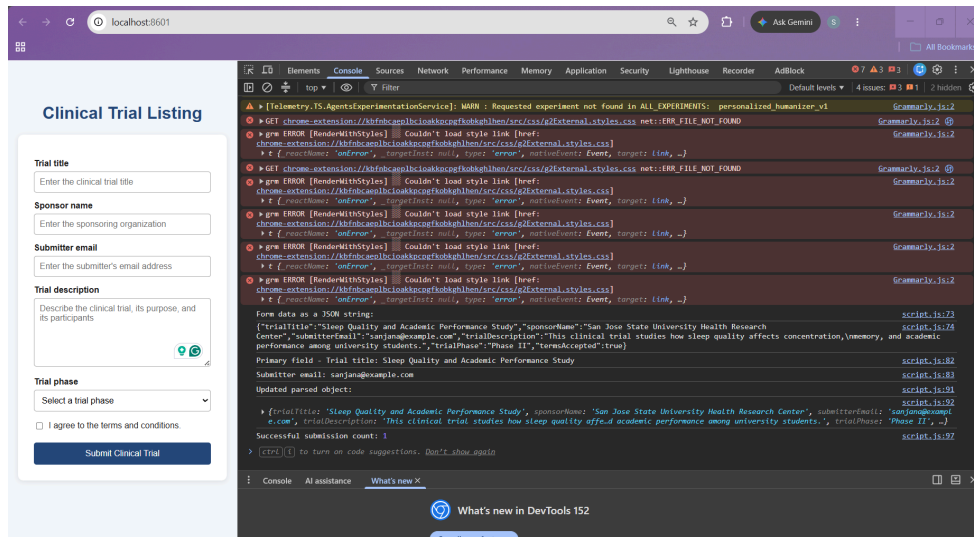


FIGURE 6: SUCCESSFUL FORM AND CONSOLE OUTPUT

## 5. LOCAL DOCKER DEPLOYMENT

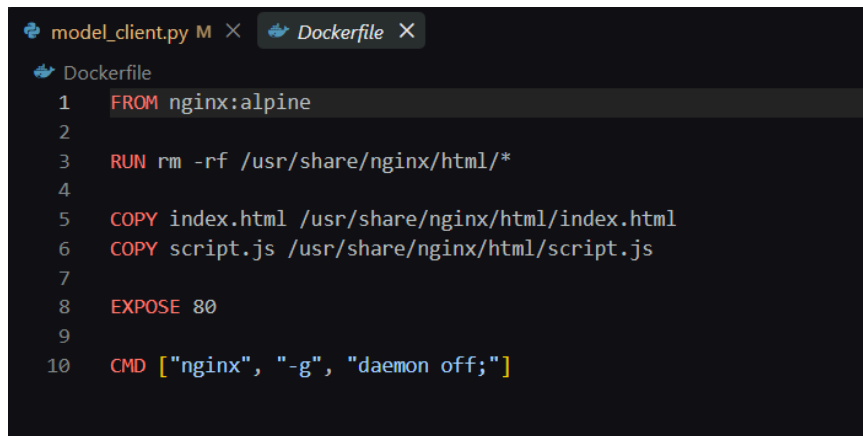


FIGURE 7: dockerfile

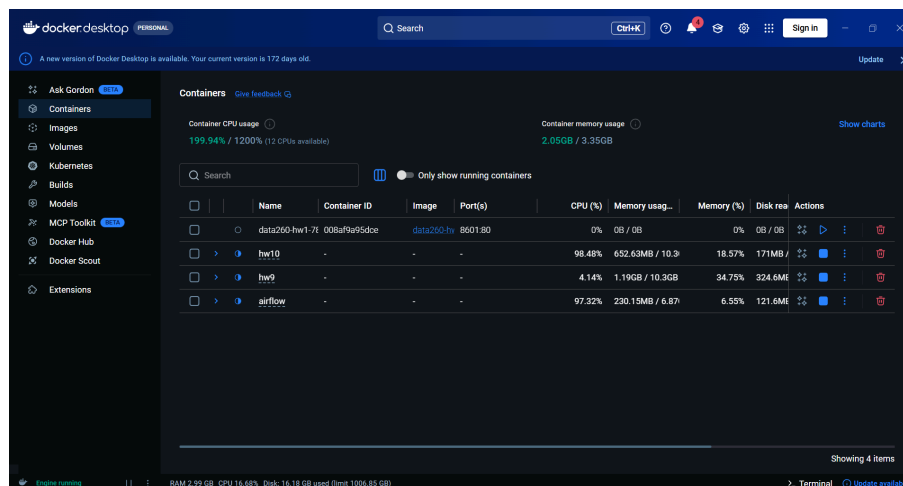


FIGURE 8: docker-ps

## 6. AWS ECS DEPLOYMENT

My student AWS credits were exhausted. The local Docker image and localhost deployment were completed and verified successfully. This limitation is reported transparently in the submitted work.

## PART 2 - AGENTIC AI

### 1. DESIGN

I implemented two model agents and one non-agent finalization step:

- Planner: proposes exactly three topical tags and a summary.

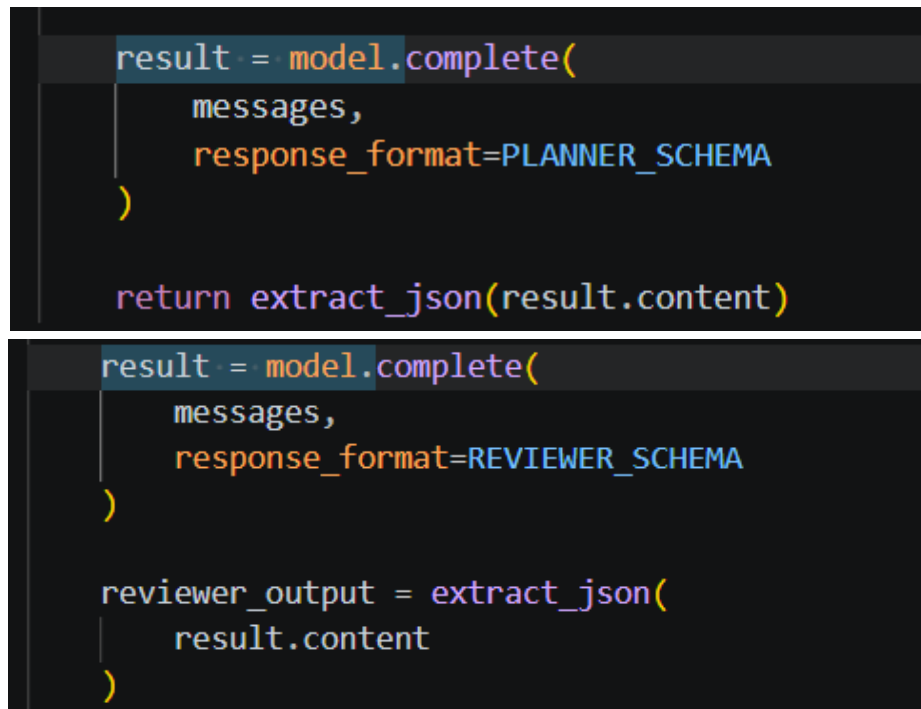
- Reviewer: checks topicality, uniqueness, and the 25-word summary limit.
- Finalizer: deterministically guarantees exactly three tags and a summary of at most 25 words.

All model calls pass through the reusable adapter in `src/model_client.py`. The agent logic does not contain fixed clinical-trial tags. Tags and summaries are derived from the supplied title and content.

## 2. COMMAND

python code/agents\_demo.py --model qwen3:4b --title "Sleep Quality and Academic Performance Study" --content "This clinical trial studies how sleep duration and sleep quality affect concentration, memory, and academic performance among university students."

## 3. PLANNER AND REVIEWER CODE



```

result = model.complete(
    messages,
    response_format=PLANNER_SCHEMA
)

return extract_json(result.content)

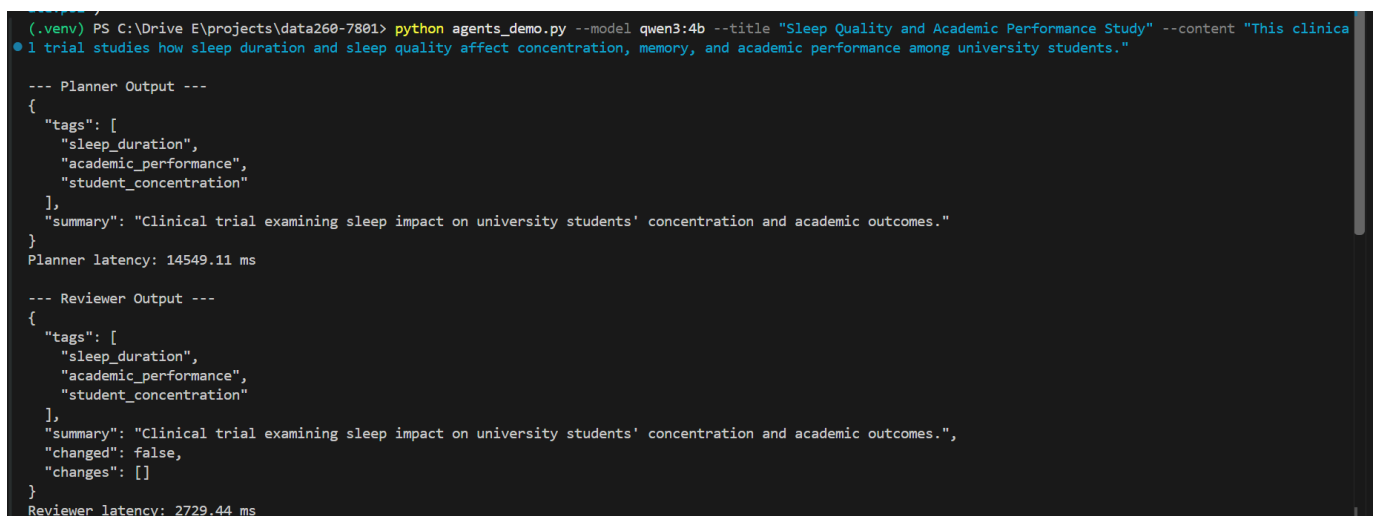
result = model.complete(
    messages,
    response_format=REVIEWER_SCHEMA
)

reviewer_output = extract_json(
    result.content
)

```

FIGURE 9: planner and reviewer code

The Reviewer's changed field is validated by directly comparing the Reviewer tags and summary with the Planner output. This prevents the model from claiming that it changed something when the visible values are identical.



```

(.venv) PS C:\Drive E\projects\data260-7801> python agents_demo.py --model qwen3:4b --title "Sleep Quality and Academic Performance Study" --content "This clinical trial studies how sleep duration and sleep quality affect concentration, memory, and academic performance among university students."

--- Planner Output ---
{
  "tags": [
    "sleep_duration",
    "academic_performance",
    "student_concentration"
  ],
  "summary": "Clinical trial examining sleep impact on university students' concentration and academic outcomes."
}
Planner latency: 14549.11 ms

--- Reviewer Output ---
{
  "tags": [
    "sleep_duration",
    "academic_performance",
    "student_concentration"
  ],
  "summary": "Clinical trial examining sleep impact on university students' concentration and academic outcomes.",
  "changed": false,
  "changes": []
}
Reviewer latency: 2729.44 ms

```

FIGURE 10: PLANNER AND REVIEWER OUTPUT

```
(.venv) PS C:\Drive E\projects\data260-7801> python agents_demo.py --model qwen3:4b --title "Sleep Quality and Academic Performance Study" --content "This clinica

--- Finalized Output ---
{
  "tags": [
    "sleep_duration",
    "academic_performance",
    "student_concentration"
  ],
  "summary": "Clinical trial examining sleep impact on university students' concentration and academic outcomes."
}

--- Publish Output ---
{
  "title": "Sleep Quality and Academic Performance Study",
  "content": "This clinical trial studies how sleep duration and sleep quality affect concentration, memory, and academic performance among university students.",
  "model": "qwen3:4b",
  "temperature": 0.0,
  "planner": {
    "tags": [
      "sleep_duration",
      "academic_performance",
      "student_concentration"
    ],
    "summary": "Clinical trial examining sleep impact on university students' concentration and academic outcomes."
  },
  "reviewer": {
    "tags": [
      "sleep_duration",
      "academic_performance",
      "student_concentration"
    ],
    "summary": "Clinical trial examining sleep impact on university students' concentration and academic outcomes.",
    "changed": false,
    "changes": []
  },
  "final": {
    "tags": [
      "sleep_duration",
      "academic_performance",
      "student_concentration"
    ],
    "summary": "Clinical trial examining sleep impact on university students' concentration and academic outcomes."
  },
  "latencyMs": 20916.93
}
```

**FIGURE 11: FINALIZED AND PUBLISHED OUTPUT**

Q1. What three final tags were obtained?

- sleep\_duration
- academic\_performance
- student\_concentration

Q2. What was the final summary?

Clinical trial examining sleep impact on university students' concentration and academic outcomes.

Q3. Did the Reviewer change anything?

No. The Reviewer determined that the Planner's three tags were topical and distinct and that its summary was already under the 25-word limit.

## 5. STEP EXPLANATION

The Planner receives the fixed title and content and creates a structured proposal. A JSON schema forces the proposal to contain exactly three tags and a summary.

The Reviewer receives the original title and content along with the Planner proposal. It checks whether the tags are topical and distinct and whether the summary follows the word limit.

The finalization step normalizes the tags, removes duplicates, fills missing tags using phrases derived from the input, and truncates the summary if it exceeds 25 words. It is a Python validation step rather than a third model agent.

The final Publish output contains the title, content, model name, temperature, Planner output, Reviewer output, final tags and summary, and measured latency.

## PART 3 - MEASURING NONDETERMINISM

### 1. FIXED INPUT

The unchanged input was saved to:  
reports/hw01/cases/nondeterminism\_input.json

```
{
  "title": "Sleep Quality and Academic Performance Study",
```

```
"content": "This clinical trial studies how sleep duration and sleep quality affect concentration, memory, and academic performance among university students."
}
```

## 2. EXPERIMENT

I ran the complete Planner-Reviewer-Finalizer pipeline:

- 20 times at temperature 0.7
- 20 times at temperature 0.0
- 40 successful runs in total

Each result contains its temperature, run number, timestamp, three tags, summary, Planner latency, Reviewer latency, total latency, and Reviewer status.

Raw results were saved as:

- reports/hw01/raw/nondeterminism\_results.json
- reports/hw01/raw/nondeterminism\_results.csv
- reports/hw01/raw/nondeterminism\_metrics.json

## 3. TAG RESULTS

| Metric                            | Temperature 0.7                                                                                                                                                                                    | Temperature 0.0                                                |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Distinct tag sets                 | 11                                                                                                                                                                                                 | 1                                                              |
| Tags in all 20 runs               | None                                                                                                                                                                                               | Academic_performance,<br>concentration_effects, sleep_duration |
| Tags appearing in exactly one run | academic_concentration;<br>concentration_impact;<br>concentration_metrics;<br>concentration_outcomes;<br>student_cognitive_function;<br>student_concentration_studies;<br>university_sleep_studies | None                                                           |

## 4. LATENCY RESULTS

| Metric | Temperature 0.7 | Temperature 0.0 |
|--------|-----------------|-----------------|
| p50    | 5526.03 ms      | 5105.43 ms      |
| p95    | 7654.37 ms      | 5529.57 ms      |
| p99    | 18532.39 ms     | 7052.89 ms      |

```
(.venv) PS C:\Drive E\projects\data260-7801> python run_nondeterminism.py
Fixed input loaded:
{
  "title": "Sleep Quality and Academic Performance Study",
  "content": "This clinical trial studies how sleep duration and sleep quality affect concentration, memory, and academic performance among university students."
}

Model: qwen3:4b
Target: 20 runs per temperature

Temperature 0.7: 20 runs already completed.

Temperature 0.0: 20 runs already completed.

=== Experiment Metrics ===
{
  "0.7": {
    "temperature": 0.7,
    "runCount": 20,
    "distinctTagSets": 11,
    "tagsInAllRuns": [],
    "tagsInExactlyOneRun": [
      "academic_concentration",
      "concentration_impact",
      "concentration_metrics",
      "concentration_outcomes",
      "student_cognitive_function",
      "student_concentration_studies",
      "university_sleep_studies"
    ],
    "latencyP50%": 5526.03,
    "latencyP99%": 7654.37,
    "latencyP99%": 18532.39
  },
  "0.0": {
    "temperature": 0.0,
    "runCount": 20,
    "distinctTagSets": 1,
    "tagsInAllRuns": [
      "academic_performance",
      "concentration_effects",
      "sleep_duration"
    ],
    "tagsInExactlyOneRun": [],
    "latencyP50%": 5105.43,
    "latencyP99%": 5529.57,
    "latencyP99%": 7052.89
  }
}

Files created:
reports\hw01\raw\nondeterminism_results.json
reports\hw01\raw\nondeterminism_results.csv
reports\hw01\raw\nondeterminism_metrics.json
```

**FIGURE12: NONDETERMINISM METRICS**

## 5. INTERPRETATION

At temperature 0.7, two users sending the same title and content may receive different but related tag sets. For example, one user might receive `sleep_quality`, `academic_performance`, and `university_students`, while another might receive `sleep_duration`, `academic_performance`, and `student_concentration`. At temperature 0.0, the same input produced the same tag set in all 20 runs. Temperature 0.0 therefore produced more repeatable results.

Variation is acceptable for brainstorming, discovery, and generating alternative topical tags because multiple descriptions may be equally useful. Variation is unacceptable when the output determines clinical-trial eligibility, participant safety, or another consequential decision. Such decisions require validated rules, consistency, and human review.

## PART 4 - MODEL CLIENT AND TOKEN ACCOUNTING

### 1. REUSABLE ADAPTER

The reusable adapter is implemented in `src/model_client.py`. Its stable interface is:

```
def complete(
    self,
    messages,
    tools=None,
    response_format=None
):
```

It sends requests to the local Ollama server and returns:

- Response content
- Input tokens
- Output tokens
- Total tokens
- Raw response metadata

All project model calls use this adapter.

### 2. AGENT.md

AGENT.md instructs the model to produce strict bullet-only code reviews. The interactive client also validates that every nonempty displayed response line begins with “- ”.

All five responses passed this verification.

### 3. TOKEN RESULTS

| Turn | Input Tokens | Output Tokens | Total Tokens | Cumulative Input | Cumulative Output | Serialized History |
|------|--------------|---------------|--------------|------------------|-------------------|--------------------|
| 1    | 134          | 46            | 180          | 134              | 46                | 249                |



|   |     |    |     |      |     |      |
|---|-----|----|-----|------|-----|------|
| 2 | 191 | 33 | 224 | 325  | 79  | 469  |
| 3 | 241 | 61 | 302 | 566  | 140 | 881  |
| 4 | 313 | 73 | 386 | 879  | 213 | 1298 |
| 5 | 403 | 53 | 456 | 1282 | 266 | 1667 |

#### 4. /stats AFTER TURN 3

Turn count: 3

Cumulative input tokens: 566

Cumulative output tokens: 140

Cumulative total tokens: 706

Serialized conversation-history length: 881 characters

```

--- Statistics after turn 3 ---
Turn count: 3
Cumulative input tokens: 566
Cumulative output tokens: 140
Cumulative total tokens: 706
Serialized conversation-history length: 881 characters

```

FIGURE 13: /stats AFTER TURN 3

#### 5. /stats AFTER TURN 5

Turn count: 5

Cumulative input tokens: 1,282

Cumulative output tokens: 266

Cumulative total tokens: 1,548

Serialized conversation-history length: 1,667 characters

```

--- Statistics after turn 5 ---
Turn count: 5
Cumulative input tokens: 1282
Cumulative output tokens: 266
Cumulative total tokens: 1548
Serialized conversation-history length: 1667 characters

```

FIGURE 14: /stats AFTER TURN 5

#### 6. REQUIRED QUESTIONS

Why is prior conversation context resent with every turn?

A model request is normally stateless. The application must resend earlier messages so the model can understand references to previous questions and answers and continue the conversation coherently.

How is a system prompt different from a user message?

A system prompt defines the model's overall role, behavioral constraints, and response format. A user message contains the specific task or question. System instructions have higher priority and remain applicable throughout the conversation.

Why do input tokens grow over a conversation?

Each request includes the system prompt, earlier user messages, earlier assistant responses, and the newest user message. As that serialized history becomes longer, the number of input tokens grows.

What eventually limits that growth?

The model's context-window limit eventually restricts the number of tokens that can be included in one request. An application must then remove old messages, summarize them, or compress the conversation history.

#### VERIFICATION

I created `verify_hw1.py` to verify:

- Required files
- Python version
- HTML requirements

- JavaScript requirements
- Model-adapter use
- Forty nondeterminism results
- Five-turn token accounting
- Bullet-only compliance

The verification output reported passed: true.

```

(.venv) PS C:\Drive E\projects\data260-7801> python verify_hw1.py
{
  "assignment": "DATA-260 Homework 1",
  "student": "Sanjana Thummalapalli",
  "sid4": 7801,
  "verifySeed": 267801,
  "timestamp": "2026-08-31T18:06:56.930100+00:00",
  "passed": true,
  "checks": [
    {
      "name": "required_files",
      "passed": true,
      "details": "All required files exist."
    },
    {
      "name": "python_version",
      "passed": true,
      "details": "3.11.0"
    },
    {
      "name": "html_requirements",
      "passed": true,
      "details": "HTML requirements passed."
    },
    {
      "name": "javascript_requirements",
      "passed": true,
      "details": "JavaScript requirements passed."
    },
    {
      "name": "model_adapter_usage",
      "passed": true,
      "details": "Agent and experiment calls use ModelClient."
    },
    {
      "name": "nondeterminism_results",
      "passed": true,

```

**FIGURE 15: VERIFICATION OUTPUT**

## REPRODUCIBLE RUN INSTRUCTIONS

Python environment:

py -3.11 -m venv .venv

.\venv\Scripts\Activate.ps1

python -m pip install -r requirements.txt

ollama pull qwen3:4b

Local Docker application:

docker build -f code/Dockerfile -t data260-hw1:latest .

docker run --name data260-hw1-7801 -d -p 8601:80 data260-hw1:latest

Agent pipeline:

python code/agents\_demo.py --model qwen3:4b --title "Sleep Quality and Academic Performance Study" --content "This clinical trial studies how sleep duration and sleep quality affect concentration, memory, and academic performance among university students."

Nondeterminism experiment:

python code/run\_nondeterminism.py

Interactive client:

python code/hw1\_client.py

Verification:

python code/verify\_hw1.py

